

long (z-independence) (can be treated as planar problem)



$$\left(-\kappa \frac{\partial T}{\partial r}(r, \theta, t) r d\theta + \kappa \frac{\partial T}{\partial r}(r+d\theta, \theta, t) (r+dr) d\theta \right)$$

radial heat inflow rate

$$\left(-\frac{\kappa}{r} \frac{\partial T}{\partial \theta}(r, \theta, t) dr + \frac{\kappa}{r} \frac{\partial T}{\partial \theta}(r, \theta+d\theta, t) dr \right) dt$$

tangential heat inflow rate

$$= \rho C r d\theta dr (T(r, \theta, t+dt) - T(r, \theta, t))$$

$$\Rightarrow \left(\kappa \frac{\partial^2 T}{\partial r^2}(r, \theta, t) r dr d\theta + \kappa \frac{\partial T}{\partial r}(r, \theta, t) dr d\theta + \frac{\kappa}{r} \frac{\partial^2 T}{\partial \theta^2}(r, \theta, t) dr d\theta \right) dt = \rho C r \frac{\partial T}{\partial t}(r, \theta, t) dr d\theta dt$$

$$\Rightarrow \alpha \left(\frac{\partial^2 T}{\partial r^2} + \frac{1}{r} \frac{\partial T}{\partial r} + \frac{1}{r^2} \frac{\partial^2 T}{\partial \theta^2} \right) = \frac{\partial T}{\partial t} \quad \forall \text{ interior points } t \geq 0$$

Given $T(r, \theta, 0) = T_0 \quad \forall \text{ interior points}$



$$\kappa \frac{\partial T}{\partial r}(R_1, \theta, t) R_1 d\theta = 0 \quad \left(\text{as tangential heat exchange is a second order infinitesimal in net} \right)$$

$$\Rightarrow \frac{\partial T}{\partial r}(R_1, \theta, t) = 0 \quad \forall \theta \in (-\pi, \pi), t \geq 0$$



$$-\kappa \frac{\partial T}{\partial r}(R_2, \theta, t) R_2 d\theta = h_i (T(R_2, \theta, t) - T_i) R_2 d\theta \quad (")$$

$$\Rightarrow h_i (T(R_2, \theta, t) - T_i) + \kappa \frac{\partial T}{\partial r}(R_2, \theta, t) = 0$$

$$h_i (T(R_2, \theta, t) - T_i) + \kappa \frac{\partial T}{\partial r}(R_2, \theta, t) = 0 \quad \forall \theta \in (0, \pi), t \geq 0$$

$$h_2 (T(R_2, \theta, t) - T_2) + \kappa \frac{\partial T}{\partial r}(R_2, \theta, t) = 0 \quad \forall \theta \in (-\pi, 0), t \geq 0$$

$$R_2 = R_1 + i \Delta r \quad \Delta r = \frac{R_2 - R_1}{N_r} \quad i \in \{0 \dots N_r\}$$

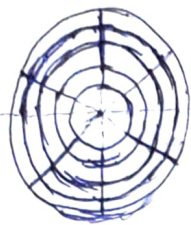
$$t_n = n \Delta t$$

$$n \in \mathbb{N}$$

$$\theta_j = (2j+1) \frac{\Delta \theta}{2}$$

$$\Delta \theta = \frac{2\pi}{N_\theta}$$

$$N_\theta \text{ is even} \quad j \in \{0 \dots N_\theta - 1\}$$



so that we always have convective BC clarity

Temperature array $T_{ij}^{(n)}$

$T_{0,-1}$	$T_{0,0}$	$T_{0,N_\theta-1}$	T_{0,N_θ}
$T_{1,-1}$	$T_{1,0}$	$T_{1,N_\theta-1}$	T_{1,N_θ}
$\vdots$	$\vdots$	$\vdots$	$\vdots$
$T_{N_r,-1}$	$T_{N_r,0}$	$T_{N_r,N_\theta-1}$	T_{N_r,N_θ}

$$T_{ij}^{(0)} = T_b$$

$$\forall i \in \{0 \dots N_r\} \quad n \geq 0$$

$$T_{i,-1}^{(n)} = T_{i,N_\theta-1}^{(n)}$$

$$T_{i,N_\theta}^{(n)} = T_{i,0}^{(n+1)}$$

The columns at the end are just to make the inner (meaning i) array circular in θ rather than linear. ($j=-1 \equiv j=N_\theta-1$, $j=N_\theta \equiv j=0$)

We use FTCS scheme for its computational ease

$$\alpha \left(\frac{T_{i+1,j}^{(n)} + T_{i-1,j}^{(n)} - 2T_{ij}^{(n)}}{(\Delta r)^2} + \frac{1}{r_i} \frac{T_{i+1,j}^{(n)} - T_{i-1,j}^{(n)}}{2\Delta r} + \frac{1}{r_i^2} \frac{T_{i,j+1}^{(n)} + T_{i,j-1}^{(n)} - 2T_{ij}^{(n)}}{(\Delta \theta)^2} \right)$$

$$= \frac{T_{ij}^{(n+1)} - T_{ij}^{(n)}}{\Delta t} + O(\Delta t, (\Delta r)^2, (\Delta \theta)^2)$$

$$\Rightarrow T_{ij}^{(n+1)} = T_{ij}^{(n)} \left(1 - \frac{2\alpha \Delta t}{(\Delta r)^2} - \frac{2\alpha \Delta t}{r_i^2 (\Delta \theta)^2} \right) + \frac{\alpha \Delta t}{r_i^2 (\Delta \theta)^2} (T_{i,j+1}^{(n)} + T_{i,j-1}^{(n)})$$

$$+ \left(\frac{\alpha \Delta t}{(\Delta r)^2} + \frac{\alpha \Delta t}{2r_i \Delta r} \right) T_{i+1,j}^{(n)} + \left(\frac{\alpha \Delta t}{(\Delta r)^2} - \frac{\alpha \Delta t}{2r_i \Delta r} \right) T_{i-1,j}^{(n)}$$

$$+ O((\Delta t)^2, (\Delta r)^2, (\Delta \theta)^2) \quad \forall i \in \{1 \dots N_r-1\}, j \in \{0 \dots N_\theta-1\}, n \geq 0$$

To estimate for $i=0$ & $i=N_r$ we use ghost points

$$\frac{\partial T}{\partial r}(R_1, \theta, t) = 0 \Rightarrow T_{-1,j}^{(n)} = T_{1,j}^{(n)} + O((\Delta r)^3) \quad \forall j \in \{0 \dots N_\theta-1\}, n \geq 0$$

Giving,

$$T_{0,j}^{(n+1)} = T_{0,j}^{(n)} \left(1 - \frac{2\alpha \Delta t}{(\Delta r)^2} - \frac{2\alpha \Delta t}{r_0^2 (\Delta \theta)^2} \right) + \frac{\alpha \Delta t}{r_0^2 (\Delta \theta)^2} (T_{0,j+1}^{(n)} + T_{0,j-1}^{(n)})$$

$$+ \frac{2\alpha \Delta t}{(\Delta r)^2} T_{1,j}^{(n)} + O(\Delta t, \Delta r, (\Delta \theta)^2) \quad \forall j \in \{0 \dots N_\theta-1\}, n \geq 0$$

Discretizing outer boundary cond.

$$h_k (T_{N_x, j}^{(n)} - T_k) + \kappa \left(\frac{T_{N_x+1, j}^{(n)} - T_{N_x-1, j}^{(n)}}{2\Delta x} \right) = O(\Delta x)^2$$

$$\Rightarrow T_{N_x+1, j}^{(n)} = T_{N_x-1, j}^{(n)} + \frac{2\Delta x h_k}{\kappa} (T_k - T_{N_x, j}^{(n)}) + O(\Delta x)^2$$

~~where~~ where, $j < \frac{N_0}{2} \Rightarrow k=1$ & $j \geq \frac{N_0}{2} \Rightarrow k=2$

$$\forall j \in \{0, \dots, N_0-1\}, n \geq 0$$

Convg,

$$\begin{aligned} T_{N_x, j}^{(n+1)} &= \frac{\alpha \Delta t}{k_{N_x}^2 (\Delta \theta)^2} (T_{N_x, j+1}^{(n)} + T_{N_x, j-1}^{(n)}) + \frac{2\alpha \Delta t}{(\Delta x)^2} T_{N_x-1, j}^{(n)} \\ &+ T_{N_x, j}^{(n)} \left(1 - \frac{2\alpha \Delta t}{(\Delta x)^2} - \frac{2\alpha \Delta t}{k_{N_x}^2 (\Delta \theta)^2} \right) \\ &+ \frac{\alpha \Delta t h_k}{\kappa} \left(\frac{2}{\Delta x} + \frac{1}{k_{N_x}} \right) (T_k - T_{N_x, j}^{(n)}) + O(\Delta x, \Delta t, (\Delta \theta)^2) \end{aligned}$$

where, $j < \frac{N_0}{2} \Rightarrow k=1$ & $j \geq \frac{N_0}{2} \Rightarrow k=2$

$$\forall j \in \{0, \dots, N_0-1\}, n \geq 0$$

For stability analysis, let the difference eqn. be equivalent to

$$T_{ij}^{(n+1)} = a_i T_{ij} + b_i (T_{i, j+1}^{(n)} + T_{i, j-1}^{(n)}) + (c + d_i) T_{i+1, j}^{(n)} + (c - d_i) T_{i-1, j}^{(n)}$$

~~Let~~

$$E_{ij}^{(n)} = \sum_{m=-\infty}^{\infty} \sum_{p=-\infty}^{\infty} G_{mp} \frac{1}{n} e^{imk_x \Delta x} e^{ipk_y \Delta y}$$

For stability, $\left| \frac{E_{ij}^{(n+1)}}{E_{ij}^{(n)}} \right| < 1 \Leftrightarrow \left| \frac{E_{mp, ij}^{(n+1)}}{E_{mp, ij}^{(n)}} \right| < 1 \quad \forall m, p$

$$\begin{aligned} \text{So, } E_{mp, ij}^{(n+1)} &= a_i E_{mp, ij}^{(n)} + b_i (E_{mp, i, j+1}^{(n)} + E_{mp, i, j-1}^{(n)}) \\ &+ (c + d_i) E_{mp, i+1, j}^{(n)} + (c - d_i) E_{mp, i-1, j}^{(n)} \end{aligned}$$

$$\Rightarrow \frac{G_{mp, n+1}}{G_{mp, n}} = a_i + b_i (e^{ipk_y \Delta y} + e^{-ipk_y \Delta y}) + (c + d_i) e^{imk_x \Delta x} + (c - d_i) e^{-imk_x \Delta x}$$

$$\text{So, } \left| \frac{G_{mp, n+1}}{G_{mp, n}} \right| < 1 \Rightarrow |a_i + 2b_i \cos(pk_y \Delta y) + 2c \cos(mk_x \Delta x) + 2c d_i \sin(mk_x \Delta x)| < 1$$

$$\Rightarrow (a_i + 2b_i \cos(pk_0 \Delta x) + 2c \cos(mk_x \Delta x))^2 + 4d_i^2 \sin^2(mk_x \Delta x) \geq 1$$

$$a_i = 1 - \frac{2\alpha \Delta t}{(\Delta x)^2} - \frac{2\alpha \Delta t}{\Delta x_i^2 (\Delta x)^2}, \quad b_i = \frac{\alpha \Delta t}{\Delta x_i^2 (\Delta x)^2}, \quad c = \frac{\alpha \Delta t}{(\Delta x)^2},$$

$$d_i = \frac{\alpha \Delta t}{2\Delta x_i \Delta x}$$

$$a_i + 2b_i + 2c = 1$$

$$\Rightarrow (2b_i(1-x) + 2c(1-y) - 1)^2 + 4d_i^2(1-y^2) < 1$$

$$x = \cos(pk_0 \Delta x), \quad y = \cos(mk_x \Delta x)$$

We need to find inequalities for a_i, b_i, c, d_i such that for all x, y independently varying b/w -1 & 1 , this inequality is satisfied

$$\Rightarrow 1 + (2b_i(1-x) + 2c(1-y))^2 - 4(b_i(1-x) + c(1-y))^2 + 4d_i^2(1-y^2) < 1$$

$$\Rightarrow d_i^2(1-y^2) < (b_i(1-x) + c(1-y))(1 - b_i(1-x) - c(1-y))$$

$$\Rightarrow d_i^2(1+y)(1-y) < (b_i(1-x) + c(1-y))(a_i + b_i(1+x) + c(1+y))$$

$$\Rightarrow d_i^2 < (b_i \left(\frac{1-x}{1-y} \right) + c) \left(c + \frac{a_i + b_i(1+x)}{1+y} \right)$$

$$\Rightarrow d_i^2 < c^2 + b_i \left(\frac{1-x}{1-y} \right) \left(c + \frac{a_i}{1+y} \right) + \frac{a_i c}{1+y} + b_i c \left(\frac{1+x}{1+y} \right) + b_i^2 \left(\frac{1-x}{1-y} \right) \left(\frac{1+x}{1+y} \right)$$

$d_i^2 < \text{minimum of RHS w.r.t. } x \& y$

if $a_i \leq 0$, then $y \rightarrow -1$ produces unboundedly $\rightarrow$ min of RHS

$$\Rightarrow a_i > 0 \Rightarrow 1 - 2b_i - 2c > 0$$

~~The same can be said if we further tighten up the limits on a_i & y in a new variable z independent of y varying b/w -1 & 1 and get min w.r.t. z and x and $z=1$~~

Now, we let x vary keeping y fixed

$$f(x, y) = c^2 + b_i \left(\frac{1-x}{1-y} \right) \left(c + \frac{a_i}{1+y} \right) + \frac{a_i c}{1+y} + b_i c \left(\frac{1+x}{1+y} \right) + b_i^2 \left(\frac{1-x}{1-y} \right) \left(\frac{1+x}{1+y} \right)$$

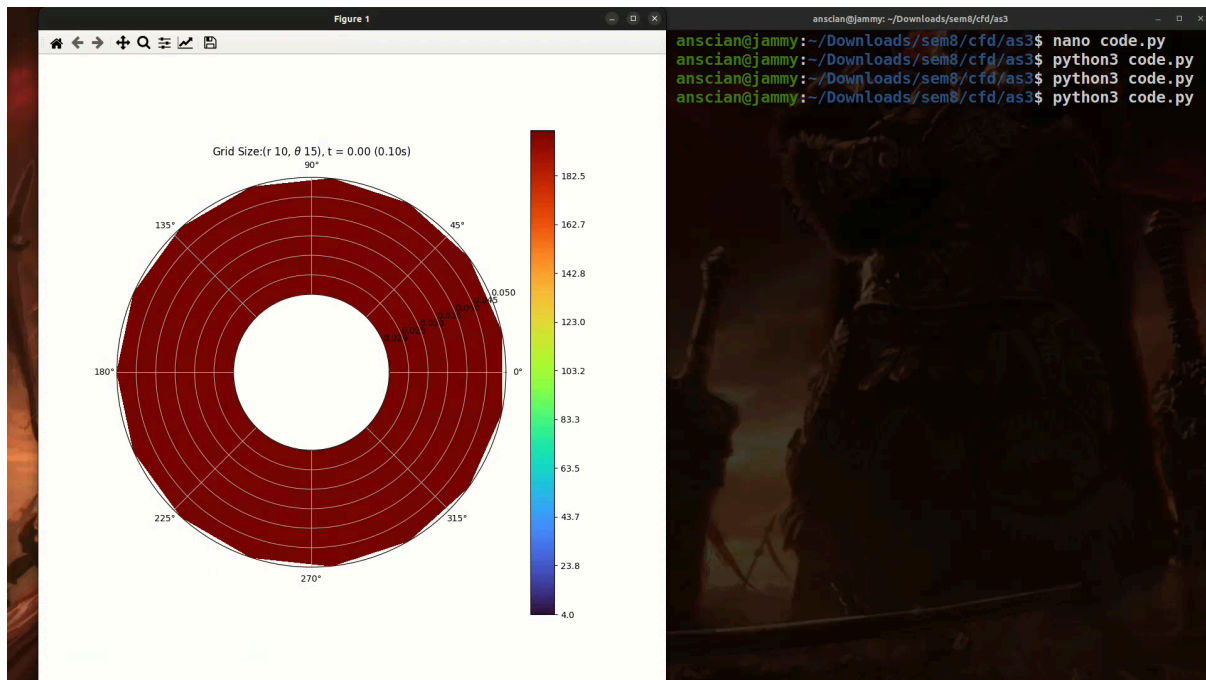
$$\frac{\partial f(x, y)}{\partial x} = -\frac{b_i}{1-y} \left(c + \frac{a_i}{1+y} \right) + \frac{b_i c}{1+y} - \frac{2b_i^2 x^*}{1-y^2} = 0$$

$$\Rightarrow x^* = \frac{1}{2b_i} (c(1-y) - c(1+y) - a_i)$$

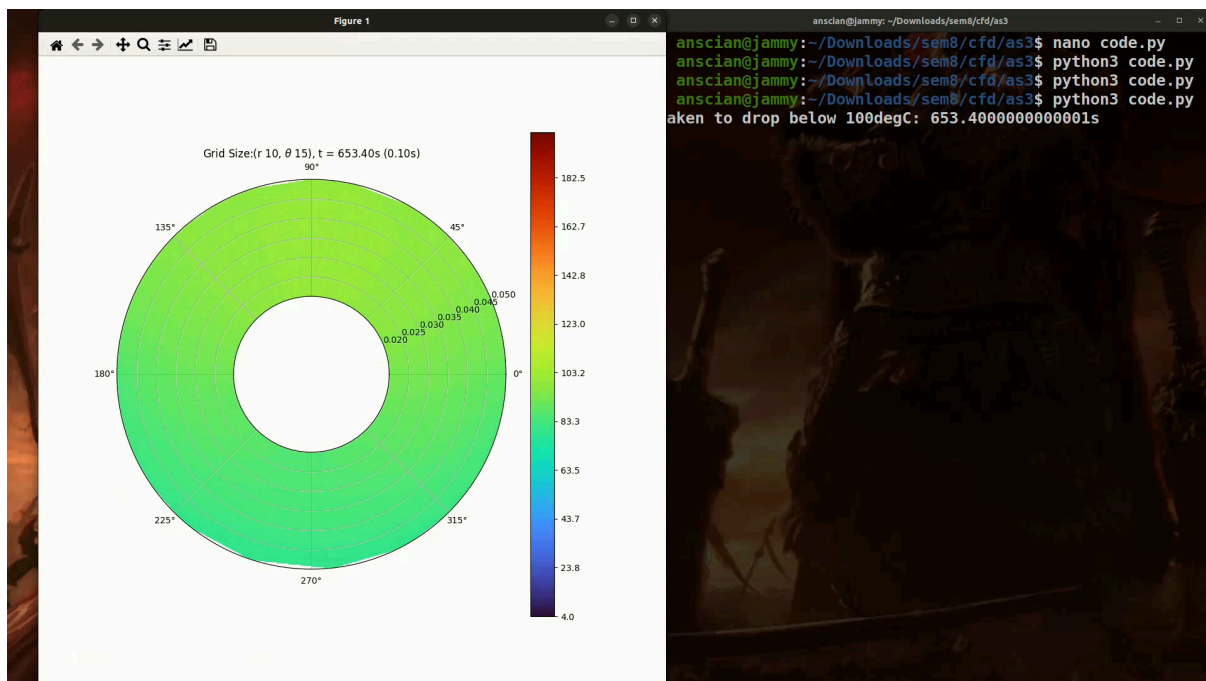
$$\Rightarrow x^* = -\frac{a_i + 2cy}{2b_i} \text{ (pt. of min)}$$

$N_r = 10$, $N_\theta = 15$, $\Delta t = 0.1s \rightarrow$ Stable Solution

Start



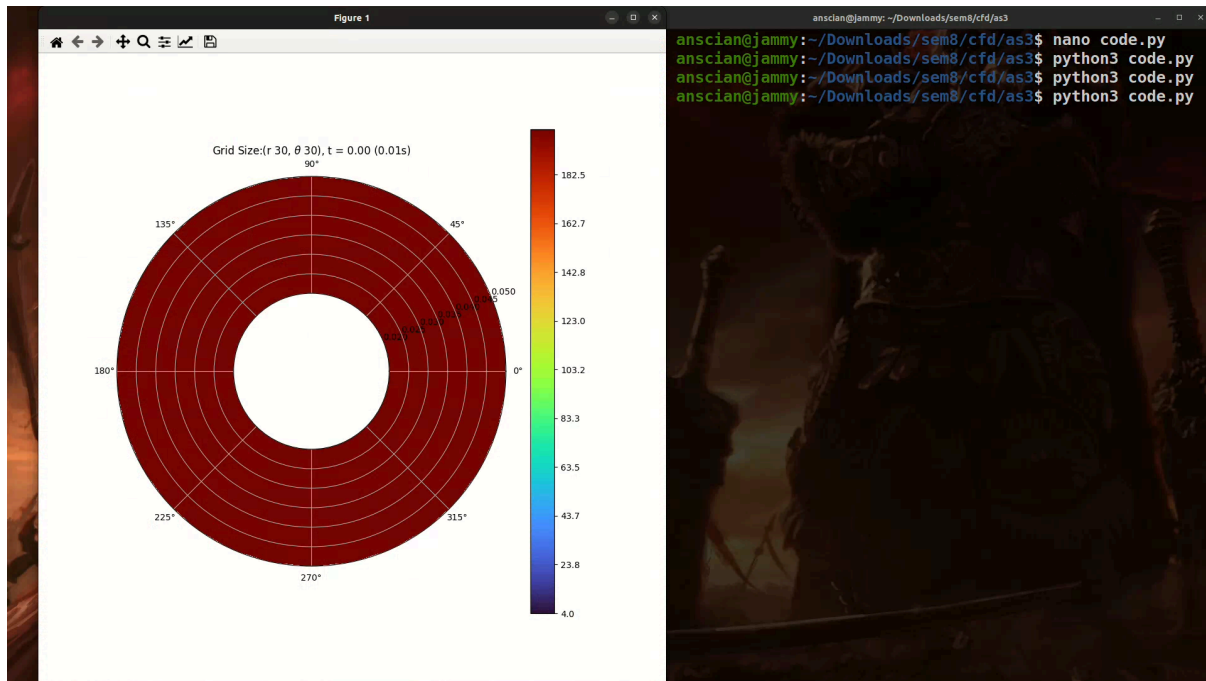
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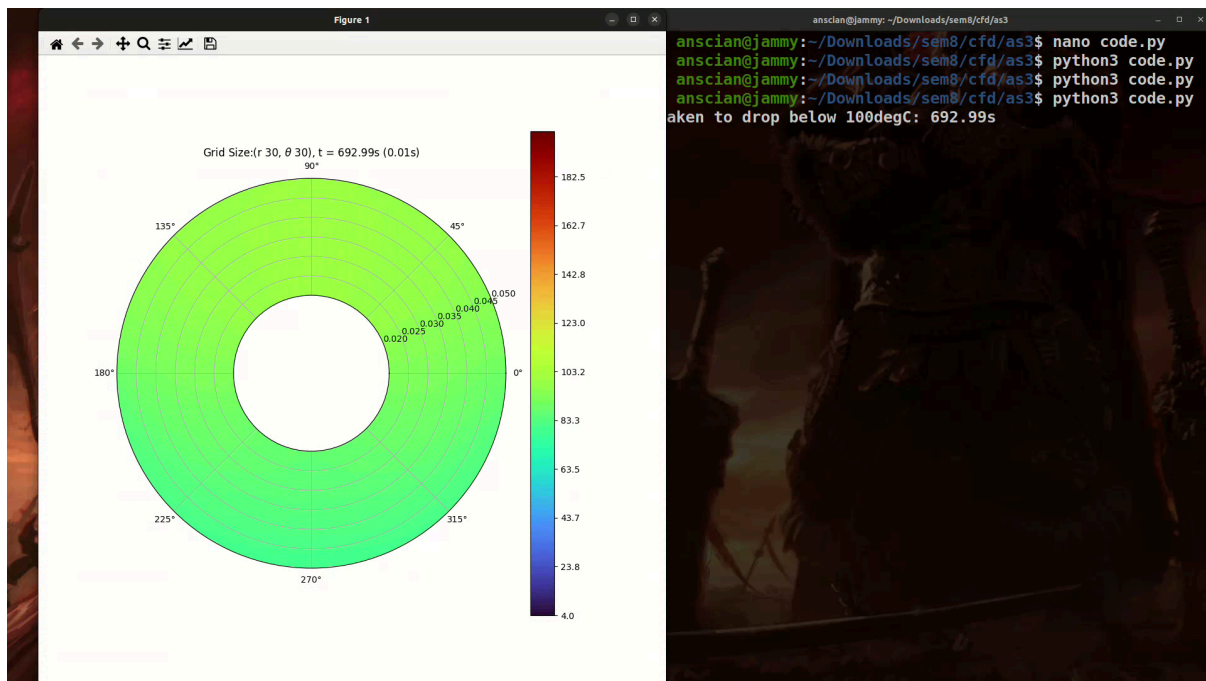
Maximum Temperature dropped below $100^{\circ}C$ after 653.40s

$N_r = 30$, $N_\theta = 30$, $\Delta t = 0.01s \rightarrow$ Stable Solution

Start



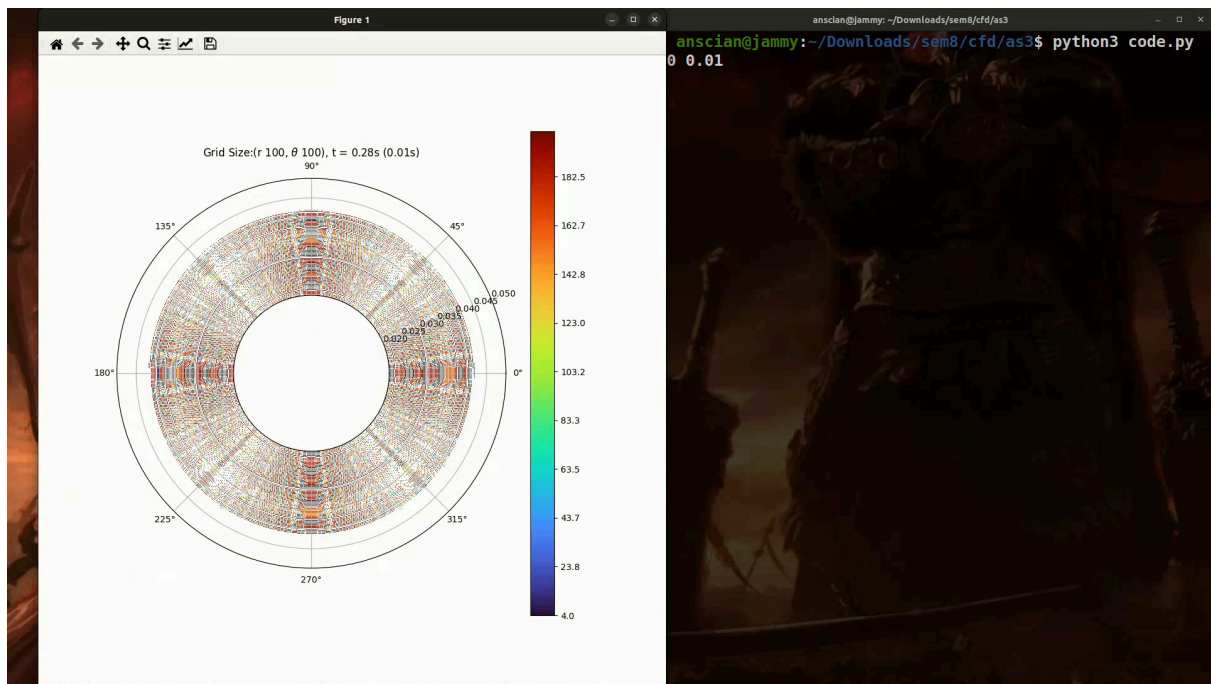
Finish



Maximum Temperature dropped below 100°C after 692.99s

$N_r = 100$, $N_\theta = 100$, $\Delta t = 0.01s \rightarrow$ Unstable Solution

Intermediate scene



White regions are where numerical overflow occurred and thus aren't plotted